COLORADO

Chief Information Officer & Executive Director

Governor's Office of Information Technology

TS-APP-010: Code Repositories (Repository Manager)

Document ID	TS-APP-010
Title	Code Repositories (Repository Manager)
Executive Sponsor	Enterprise Applications, Director of Enterprise Applications
Issuing Authority	Governor's Office of Information Technology (OIT)
Applies to	OIT and OIT-supported agencies, including employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders.
Instrument Type	Standard
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1. Purpose

OIT is statutorily obligated to provide IT support and services to the State and its stakeholders as effectively and efficiently as possible.

This document defines the standard methods that Enterprise Applications shall use to support delivery of Version Control (code repositories, configuration repositories, project artifact stratification, role-based access utilizing Active Directory, and state authentication constructs) capabilities through the adoption, use, and maintenance of an in-house code repository.

This technical standard is intended to be used as a comparator against proposed future Repository Managers. A rational determination of a specific Repository Manager will enable more efficient support services and security, and provide additional stability across development teams while improving consistency of operational efforts.

2. Scope

These standards apply across OIT and agencies supported by OIT. This standard shall be used by employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient. This is applicable to new solutions requested and major changes to existing solutions, developed by OIT staff as well as vendors developing solutions for which OIT will provide day-forward maintenance and support.

This document is intended for Enterprise Applications, but applies to any OIT unit providing these services to customers.

3. Roles and Responsibilities

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews/approves the Standard (this document).

Enterprise Architecture - Chief Enterprise Architect

- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Enterprise Applications - Director of Enterprise Applications

- Owner of Standard.
- Engages stakeholders and subject matter experts to determine the standard.
- Updates document standards and version number.
- Presents to the Architecture Review Board for approval and adoption.

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

4. Standards

Version Control Requirements

Repositories must be provided by a current OIT Certified Standard Product (i.e., Git) for Repository Managers (see TS-APP-009) and support the following:

- Manage the change and configuration of applications, including versioning, and a code/configuration repository.
- Enable a collaborative environment where multiple developers simultaneously contribute to projects.
- Allow multiple users to check out files for alteration without conflict.
- Allow various versions of files to be compared.
- Allow independent code versions if differentiated development efforts are required.
- Provide the ability to archive successive versions of source-controlled elements.
- Maintain detailed historical and version information of interactions from all users.
- Provide functionality to automate frequent tasks.
- Allow for restoration of previous versions as needed.
- Provide functionality to integrate with OIT standard Continuous Integration (CI) toolchain(s).
- Be able to scale to encompass managing multiple projects across multiple agencies.

Security Verifiable

Any Repository Manager considered for use in development, deployments, or any aspect supported by OIT and/or its vendors is required to be compatible and compliant with security

scanning and assurance tools. All Repository Managers are subject to vulnerability scanning, assessment, and mitigation prior to approval for deployment to a production environment.

Portability

Repository Managers should be portable due to the collaborative nature of the tools and may be hosted across many platforms, infrastructures, and architectures. Interaction with the Repository Manager shall be system-agnostic.

Security Frameworks

Repository Managers should be configurable to prevent common security vulnerabilities such as unauthenticated API access, remote code execution, local file inclusion, and unauthorized access to private repositories. The system and any supporting or dependent applications must comply with the level of security specified in OIT's Information Security Policies.

Accelerators

Repository Managers should offer accelerators such as templated hierarchical structures, scheduled job automation features, and a variety of webhooks for use in rapid development and delivery of versioning solutions.

Sustainability

Products and services may be considered sustainable when they are rationally accounted for from both a budgetary and staff support perspective. Budget sustainability means licensing, maintenance, and other financial commitments necessary for continued operation have been planned for and documented. Staff sustainability means ongoing support has been planned for from a lifecycle management, security, patching, and support/KLO perspective.

Authentication

- The system should be capable of supporting authentication with all major OIT directory services, including Google and Microsoft platforms.
- Should allow integration with OIT access control features to safeguard sensitive repositories.
- Should allow project-based access control via Role-Based Access Controls (RBAC).
- Should have the ability to configure and audit version changes to code or artifacts by logging the activities of all users of the repository.

5. Variance and Exception Process

Requests to use Repository Managers other than those aligned with this standard, and approved via a documented technical specification, shall be submitted in writing to the Director of Enterprise Applications and/or the CISO's Security Architecture team for possible consideration, using the standard exception form.

6. References

- OIT and Information Security Policies (see OIT Plaza - Policies): This document establishes operational standards for OIT. All staff following these standards shall do so in compliance with all other OIT policies as published. In the event of any perceived conflict, OIT policies shall prevail.

7. Definitions

- Repository Manager: A component of a Version Control System (Repo) functioning as a centralized, stable, and reliable environment used in versioning differing iterations of files or artifacts, used for source code version control as part of the Software Development Life-Cycle (SDLC) to organize, track, backup, and revert incremental steps in a development project as needed.
- Source Code: The version of software as it is originally written by a human in plain text (human-readable alphanumeric characters).
- Artifacts: Tangible products or byproducts produced during the development of software, such as libraries, existing executable code (e.g., a dynamic-linked library), or other items necessary to the overall function or execution of the source code.